Singular Transformations of Probability Density Functions

Author: *R. Scott McIntire*

Date: Aug 11, 2026

Overview

Suppose you know the joint density of a pair of random variables, (X_1, X_2). What is the density of their sum, $X_1 + X_2$? Every probability student knows the answer – the convolution formula – but it is worth asking where that formula comes from, because the map $(x_1, x_2) \mapsto x_1 + x_2$ is *singular*: it sends two dimensions to one, so the standard change-of-variables formula, which requires an invertible map, does not apply. What machine produces the convolution formula – and will the same machine answer the harder questions: the density of X_1/X_2? of $X_1^2 + \cdots + X_n^2$?

For non-singular transformations of variables with distributions, there are standard formulas that describe the probability distribution of the target variable. However, in the case of singular transformations, there is no immediate formula. Often what is done is to add independent variables in a somewhat ad hoc fashion to make the transformation non-singular, and then "integrate out" – if possible – the independent variables from the resulting formula; we exhibit this maneuver after Example 1 below, so its reliance on a lucky choice of invented variables can be seen next to the systematic alternative. Below we describe a procedure to determine the target distribution with a more systematic process.

Singular Transformation Density Formula

We state and derive a theorem for constructing the probability density function of a random variable Y which is a singular function of a random variable X. By singular, we mean that $Y = f(X)$ with f a function such that $f : \mathbf{R}^n \to \mathbf{R}^m$ and $m < n$.

Theorem 1. *Let $f : \mathcal{D} \subset \mathbf{R}^n \to \mathbf{R}^m$ be a continuously differentiable function with $n > m$, such that $Df(\mathbf{x})$ has rank m for all $\mathbf{x} \in \mathcal{D}$, and let P_X be a continuous probability density function of the random variable X which vanishes outside $\mathcal{D}$. Then each nonempty level set $f^{-1}(\mathbf{y})$ is an $(n-m)$-dimensional C^1 manifold, and the random variable $Y = f(X)$ has a density P_Y given for almost every $\mathbf{y}$ by the surface integral*

$$P_Y(\mathbf{y}) = \int_{f^{-1}(\mathbf{y})} \frac{P_X(\mathbf{x})}{\sqrt{|Df(\mathbf{x})\, Df^T(\mathbf{x})|}}\, dS \tag{1}$$

where dS denotes $(n-m)$-dimensional surface measure on $f^{-1}(\mathbf{y})$. If, in addition, the integral in (1) converges locally uniformly in $\mathbf{y}$, then this version of P_Y is continuous and (1) holds for every $\mathbf{y}$.

For computation we use a parameterized form of (1). If $f^{-1}(\mathbf{y})$ is covered – up to a set of surface measure zero – by $k(\mathbf{y})$ disjoint injective C^1 parameterizations $\mathbf{x}_i(\mu; \mathbf{y})$, $i \in [1, k(\mathbf{y})]$, $\mu \in \Omega_i \subset \mathbf{R}^{n-m}$, then (1) becomes

$$P_Y(\mathbf{y}) = \sum_{i=1}^{k(\mathbf{y})} \int_{\Omega_i} \frac{P_X(\mathbf{x}_i(\mu;\mathbf{y}))}{\sqrt{|Df(\mathbf{x}_i(\mu;\mathbf{y}))\ Df^T(\mathbf{x}_i(\mu;\mathbf{y}))|}} \sqrt{\left|D_\mu \mathbf{x}_i^T(\mu;\mathbf{y})\ D_\mu \mathbf{x}_i(\mu;\mathbf{y})\right|}\, d\mu \tag{2}$$

Formula (1) is a form of the smooth coarea formula of geometric measure theory [1, 2]. The aim of this note is a self-contained derivation and a collection of worked examples.

The theorem is best understood as the change-of-variables formula the reader already knows, played on a bigger board. The dictionary:

Non-singular $(m = n)$	**Singular** $(m < n)$
preimage of $\mathbf{y}$: a single point	preimage of $\mathbf{y}$: an $(n-m)$-dimensional manifold
Jacobian factor $\lvert\det Df(\mathbf{x})\rvert$	Gram factor $\sqrt{\lvert Df(\mathbf{x})\, Df^T(\mathbf{x})\rvert}$
the inverse map f^{-1}	the pseudo inverse $Df^T(Df\, Df^T)^{-1}$
$P_Y(\mathbf{y}) = P_X(f^{-1}(\mathbf{y}))/\lvert\det Df\rvert$	$P_Y(\mathbf{y}) = \int_{f^{-1}(\mathbf{y})} P_X/\sqrt{\lvert Df\, Df^T\rvert}\, dS$
change of variables	coarea formula

Each entry on the right reduces to the one on the left when $m = n$: the level set collapses to a point, the fiber integral to a single evaluation, the Gram determinant to $|\det Df|$, and the pseudo inverse to the inverse.

Prerequisite Results

We recall how to compute the pseudo inverse of a matrix which maps from a high dimension to a low dimension.

Lemma 1. *Let $A : \mathbf{R}^n \to \mathbf{R}^m$ be a linear transformation with $n > m$. If the rank of A is m, then the pseudo inverse of A is $A^T(AA^T)^{-1}$.*

Proof. A given $\mathbf{x} \in \mathbf{R}^n$ can be written uniquely as $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2$ where $\mathbf{x}_1 \in R(A^T)$ and $\mathbf{x}_2 \in N(A)$. For a given $\mathbf{y} \in \mathbf{R}^m$, it is easy to show that there is a unique $\mathbf{x}_p \in R(A^T)$ such that $A\mathbf{x}_p = \mathbf{y}$. The set of all $\mathbf{x} \in \mathbf{R}^n$ such that $A\mathbf{x} = \mathbf{y}$ has the form $\{\mathbf{x}_p + \mathbf{x}_n \mid \mathbf{x}_n \in N(A)\}$. The pseudo inverse of A is the map that sends $\mathbf{y}$ to $\mathbf{x}_p$. Since $R(A^T) \perp N(A)$, $\mathbf{x}_p$ is the solution to the problem:

$$\begin{aligned} &\min_{\mathbf{x}} \|\mathbf{x}\|^2 \\ \text{subject to: } &\mathbf{y} = A\mathbf{x} \end{aligned}$$

The solution of this can be found using Lagrange multipliers. The resulting minimization problem is:

$$\min_{\mathbf{x}} F(\mathbf{x}) = \min_{\mathbf{x}} \|\mathbf{x}\|^2 + \langle \lambda, \mathbf{y} - A\mathbf{x}\rangle$$

At a minimum, $\mathbf{x}^*$, $DF(\mathbf{x}^*)(\mathbf{h}) = \mathbf{0}$. $DF(\mathbf{x}^*)$ can be computed from $F(\mathbf{x}+\mathbf{h}) = F(\mathbf{x}) + DF(\mathbf{x})(\mathbf{h}) + o(\mathbf{h})$. The order $\mathbf{h}$ terms are:

$$2\langle \mathbf{x}, \mathbf{h}\rangle - \langle \lambda, A\mathbf{h}\rangle$$

Therefore, $DF(\mathbf{x}^*)(\mathbf{h}) = \langle 2\mathbf{x}^* - A^T\lambda, \mathbf{h}\rangle$. Since $DF(\mathbf{x}^*) \equiv 0$, we have that $2\mathbf{x}^* - A^T\lambda = \mathbf{0}$. Applying A to this last equation and using the fact that A has rank m and $A\mathbf{x}^* = \mathbf{y}$ yields:[1] $\lambda = 2(AA^T)^{-1}\mathbf{y}$. This implies

[1] Since A has rank m ($\dim(R(A)) = m$) and $R(AA^T) = R(A)$, AA^T is invertible.

that $\mathbf{x}^* = A^T \left(AA^T\right)^{-1}\mathbf{y}$. Since $\|\mathbf{x}\|^2$ is strictly convex and the constraint set is affine, this stationary point is indeed the unique minimum.

Note the shape of this argument: $\mathbf{x}_p$ is a *best approximation* – the point of the affine solution set nearest the origin – found through the orthogonal decomposition $R(A^T) \perp N(A)$. The pseudo inverse is a projection construction, the same "best approximation in a subspace" game that underlies least squares – and, in function space, the conditional expectation of probability theory.

We need the following change of volume formula:

Lemma 2. *Let $A : \mathbf{R}^n \to \mathbf{R}^m$ be a linear transformation with $n < m$, then if A has rank n it transforms the n-dimensional unit cube in $\mathbf{R}^n$ to a region in $\mathbf{R}^m$ with n-dimensional volume $\sqrt{|A^TA|}$.*

Proof. Since $R(A)$ has dimension n, there is a linear map $P : R(A) \to \mathbf{R}^n$ which is a linear isomorphism that preserves the inner product. That is, $\langle P\mathbf{x}, P\mathbf{y}\rangle = \langle \mathbf{x}, \mathbf{y}\rangle$ for all $\mathbf{x}, \mathbf{y} \in R(A)$, equivalently $\langle (P^TP - I)\mathbf{x}, \mathbf{y}\rangle = 0 \ \forall \mathbf{x}, \mathbf{y} \in R(A)$. This says P^TP acts as the identity on $R(A)$; in particular, $P^TPA = A$ since each column of A lies in $R(A)$. The change in volume of the unit cube under PA is the same as under A since P preserves volumes. However, PA is a map from $\mathbf{R}^n$ to $\mathbf{R}^n$, so it changes the volume of the unit cube by $|PA|$. Since for given transformations $B, C : \mathbf{R}^n \to \mathbf{R}^n$ we have $|B^T| = |B|$ and $|BC| = |B|\,|C|$, it follows that $|PA|^2 = |PA|\,|PA| = |A^TP^T|\,|PA| = |A^TP^TPA| = |A^TA|$, where the last equality uses $P^TPA = A$. Therefore, the change in volume is $\sqrt{|A^TA|}$.

Derivation of the Singular Transformation Formula

We now derive Theorem 1. The argument is geometric and informal at two points: the product decomposition near a level set is only local, and the linearization step is not carried out in full detail. A fully rigorous proof may be based on the coarea formula [1, 2].

Derivation. Let ν be the induced probability measure defined by[2]

$$\nu(\Omega_Y) = \int_{f^{-1}(\Omega_Y)} P_X(\mathbf{x})\, d\mathbf{x}$$

Given an arbitrary Borel measurable region Ω_Y and letting $\Omega_X = f^{-1}(\Omega_Y)$ we have that

$$\nu(\Omega_Y) = \int_{\Omega_X} P_X(\mathbf{x})\, d\mathbf{x} \tag{3}$$

Assume that for each $\mathbf{y} \in \Omega_Y$, $f^{-1}(\mathbf{y})$ is a single connected $(n-m)$-dimensional manifold (i.e., $k(\mathbf{y}) = 1$). Then there are local coordinates $\mathbf{x}_1, \mathbf{x}_2$, and manifolds Ω_{X_1}, Ω_{X_2}, such that Ω_X is a product of Ω_{X_1} and Ω_{X_2} with the component manifolds Ω_{X_1}, Ω_{X_2}, locally tangent to $R(Df^T(\mathbf{x}))$ and $N(Df(\mathbf{x}))$ respectively. Therefore,

$$\nu(\Omega_Y) = \int_{\Omega_X} P_X(\mathbf{x}(\mathbf{x}_1, \mathbf{x}_2))\, d\mathbf{x}_1\, d\mathbf{x}_2.$$

[2] The measure extends in a natural way to all Lebesgue measurable sets.

Since for a given transformation A, $R(A^T) \perp N(A)$, it follows that the product measure $d\mathbf{x}_1\, d\mathbf{x}_2$ is equal to the product of the component measures. Thus, we may write (3) as

$$\nu(\Omega_Y) = \int_{\Omega_{X_1}} \int_{\Omega_{X_2}} P_X(\mathbf{x}(\mathbf{x}_1, \mathbf{x}_2))\, d\mathbf{x}_1\, d\mathbf{x}_2.$$

For a given $\mathbf{x}_2$, $\mathbf{y}$ maps to the manifold Ω_{X_1} in an invertible way (locally). The linear approximation to the map f is Df – the derivative playing its role as *best linear approximation*; by Lemma 1 its pseudo inverse is $Df^T(Df\, Df^T)^{-1}$, so that locally

$$\mathbf{x}_1 \approx Df^T(\mathbf{x}) \left(Df(\mathbf{x})\, Df^T(\mathbf{x})\right)^{-1} \mathbf{y}.$$

The map carrying $\mathbf{y}$ to $\mathbf{x}_1$ is thus (locally) the linear map $B = Df^T \left(Df\, Df^T\right)^{-1}$, and Lemma 2 prices its volume distortion:

$$\sqrt{|B^T B|} = \sqrt{\left|(Df\, Df^T)^{-1}\, Df\, Df^T\, (Df\, Df^T)^{-1}\right|} = \sqrt{\left|(Df\, Df^T)^{-1}\right|} = \frac{1}{\sqrt{|Df\, Df^T|}}$$

– and there is the divisor of formula (1), born as a Gram determinant. Using $\mathbf{y}$ to express $\mathbf{x}_1$ we have

$$\nu(\Omega_Y) = \int_{\Omega_Y} \int_{\Omega_{X_2}} \frac{P_X(\mathbf{x}(\mathbf{y}, \mathbf{x}_2))}{\sqrt{|Df(\mathbf{x}(\mathbf{y}, \mathbf{x}_2)) Df^T(\mathbf{x}(\mathbf{y}, \mathbf{x}_2))|}}\, d\mathbf{x}_2\, d\mathbf{y}.$$

The integral above exhibits a Lebesgue density for ν, so ν is absolutely continuous with respect to Lebesgue measure[3] and there exists a measurable function $P_Y(\mathbf{y})$ such that $\nu(\Omega_Y) = \int_{\Omega_Y} P_Y(\mathbf{y})\, d\mathbf{y}$. So that

$$\int_{\Omega_Y} P_Y(\mathbf{y})\, d\mathbf{y} = \int_{\Omega_Y} \int_{\Omega_{X_2}} \frac{P_X(\mathbf{x}(\mathbf{y}, \mathbf{x}_2))}{\sqrt{|Df(\mathbf{x}(\mathbf{y}, \mathbf{x}_2)) Df^T(\mathbf{x}(\mathbf{y}, \mathbf{x}_2))|}}\, d\mathbf{x}_2\, d\mathbf{y}$$

Or,

$$\int_{\Omega_Y} \left(P_Y(\mathbf{y}) - \int_{\Omega_{X_2}} \frac{P_X(\mathbf{x}(\mathbf{y}, \mathbf{x}_2))}{\sqrt{|Df(\mathbf{x}(\mathbf{y}, \mathbf{x}_2)) Df^T(\mathbf{x}(\mathbf{y}, \mathbf{x}_2))|}}\, d\mathbf{x}_2 \right) d\mathbf{y} = 0$$

Since this is true for all Borel measurable sets, Ω_Y, the two integrands agree for almost every $\mathbf{y}$:

$$P_Y(\mathbf{y}) = \int_{\Omega_{X_2}} \frac{P_X(\mathbf{x}(\mathbf{y}, \mathbf{x}_2))}{\sqrt{|Df(\mathbf{x}(\mathbf{y}, \mathbf{x}_2)) Df^T(\mathbf{x}(\mathbf{y}, \mathbf{x}_2))|}}\, d\mathbf{x}_2 \tag{4}$$

But in this case the component manifold, Ω_{X_2}, is just $f^{-1}(\mathbf{y})$. If we parameterize this as $\mathbf{x}(\mu; \mathbf{y})$, then by Lemma 2 we have

$$P_Y(\mathbf{y}) = \int_{f^{-1}(\mathbf{y})} \frac{P_X(\mathbf{x}(\mu; \mathbf{y}))}{\sqrt{|Df(\mathbf{x}(\mu; \mathbf{y})) Df^T(\mathbf{x}(\mu; \mathbf{y}))|}} \sqrt{|D_\mu \mathbf{x}^T(\mu; \mathbf{y}) D_\mu \mathbf{x}(\mu; \mathbf{y})|}\, d\mu$$

which is (2) with $k(\mathbf{y}) = 1$.

In the general case, the level set may have several components, or may require several charts to cover it, as with $y = x_1^2 + x_2^2$ (Example 3 below), where two parameterizations together cover the circle $f^{-1}(y)$. The contributions of the charts add, giving the sum in (2).

[3] The existence of the density is the Radon–Nikodym theorem. Since Lebesgue measure is regular, the identity extends from Borel sets to all Lebesgue measurable sets.

Finally, when the integrals in (1) and (2) converge locally uniformly in $\mathbf{y}$, they define a continuous function of $\mathbf{y}$, and we may take this version as P_Y; this gives the continuity assertion of Theorem 1.

Applications of Theorem 1

We apply Theorem 1 with various types of singular mappings.

Example 1: $y = f(\mathbf{x}) = x_1 + x_2 \quad (f : \mathbf{R}^2 \to \mathbf{R})$

The set $f^{-1}(y)$ can be parameterized as $\mathbf{x}_1(\mu; y) = (\mu, y - \mu)^T$ with $\mu \in (-\infty, \infty)$.
$\sqrt{|D_\mu \mathbf{x}_1^T(\mu; y)\ D_\mu \mathbf{x}_1(\mu; y)|} = \|(1, -1)\| = \sqrt{2}$.
Also, $\sqrt{|Df(\mathbf{x}_1(\mu; y))\ Df^T(\mathbf{x}_1(\mu; y))|} = \|\nabla_\mathbf{x} f(\mathbf{x}_1(\mu; y))\| = \|(1, 1)\| = \sqrt{2}$.
Therefore, by Theorem 1 we have:

$$P_Y(y) = \int_{-\infty}^{\infty} P_{X_1 X_2}(\mu, y - \mu)\, d\mu$$

This is the familiar convolution formula: the new machine reproduces the one answer the reader already owns – which is how one learns to trust a new tool. It is also the place to exhibit the "ad hoc" route mentioned in the Overview. There, one adjoins an invented variable, say $y_2 = x_2$, so that the augmented map $(x_1, x_2) \mapsto (x_1 + x_2,\, x_2)$ is non-singular with Jacobian determinant 1; the standard change-of-variables formula gives $P_{YY_2}(y, y_2) = P_{X_1X_2}(y - y_2,\, y_2)$; and integrating out y_2 recovers the same convolution integral. The maneuver works – here – because $y_2 = x_2$ happens to be a convenient choice, and because the marginalization happens to be tractable. Theorem 1 asks for no invented variables and no luck.

Example 2: $y = f(\mathbf{x}) = x_1/x_2 \quad (f : \mathbf{R}^2 \to \mathbf{R})$

Here the natural domain is $\mathcal{D} = \{\mathbf{x} \in \mathbf{R}^2 \mid x_2 \neq 0\}$. The set $f^{-1}(y)$ can be parameterized as $\mathbf{x}_1(\mu; y) = (y\mu, \mu)^T$ with $\mu \in (-\infty, 0) \cup (0, \infty)$. We label the first and second components of $\mathbf{x}_1$ as $x_{1,1}$ and $x_{1,2}$ respectively.
$\sqrt{|D_\mu \mathbf{x}_1^T(\mu; y)\ D_\mu \mathbf{x}_1(\mu; y)|} = \|(y, 1)\| = \sqrt{1 + y^2}$. And,

$$\begin{aligned}\sqrt{|Df(\mathbf{x}_1(\mu; y))\ Df^T(\mathbf{x}_1(\mu; y))|} &= \|\nabla_\mathbf{x} f(\mathbf{x}_1(\mu; y))\| \\ &= \|(1/x_{1,2}(\mu; y), -x_{1,1}(\mu; y)/x_{1,2}(\mu; y)^2)\| \\ &= \sqrt{1 + y^2}/|\mu|\end{aligned}$$

Therefore, Theorem 1 gives:

$$P_Y(y) = \int_{-\infty}^{\infty} |\mu|\, P_{X_1X_2}(y\mu, \mu)\, d\mu$$

Example 3: $y = f(\mathbf{x}) = x_1^2 + x_2^2 \quad (f : \mathbf{R}^2 \to \mathbf{R})$

The set $f^{-1}(y)$ is the union of the parameterizations: $\mathbf{x}_1(\mu; y) = (\mu, \sqrt{y - \mu^2})^T$
and $\mathbf{x}_2(\mu; y) = (\mu, -\sqrt{y - \mu^2})^T$ with $\mu \in [-\sqrt{y}, \sqrt{y}]$. (The two charts overlap only at the endpoints $\mu = \pm\sqrt{y}$, a set of surface measure zero, as permitted by Theorem 1.)

$\sqrt{|D_\mu \mathbf{x}_1^T(\mu; y)\ D_\mu \mathbf{x}_1(\mu; y)|} = \|(1, -\mu/\sqrt{y-\mu^2})\| = \sqrt{y/(y-\mu^2)}$.

$\sqrt{|D_\mu \mathbf{x}_2^T(\mu; y)\ D_\mu \mathbf{x}_2(\mu; y)|} = \|(1, \mu/\sqrt{y-\mu^2})\| = \sqrt{y/(y-\mu^2)}$.

$\sqrt{|Df(\mathbf{x}_1(\mu; y))\ Df^T(\mathbf{x}_1(\mu; y))|} = 2\|(\mu, \sqrt{y-\mu^2})\| = 2\sqrt{y}$.

$\sqrt{|Df(\mathbf{x}_2(\mu; y))\ Df^T(\mathbf{x}_2(\mu; y))|} = 2\|(\mu, -\sqrt{y-\mu^2})\| = 2\sqrt{y}$.

Therefore, Theorem 1 gives:

$$P_Y(y) = \begin{cases} \int_{-\sqrt{y}}^{\sqrt{y}} \frac{P_{X_1X_2}(\mu, \sqrt{y-\mu^2}) + P_{X_1X_2}(\mu, -\sqrt{y-\mu^2})}{2\sqrt{y-\mu^2}}\, d\mu & y \geq 0; \\ 0 & \text{otherwise.} \end{cases}$$

Example 4: $y_1 = x_1 + x_2 + x_3$; $y_2 = x_1 - x_2$ $\quad (f : \mathbf{R}^3 \to \mathbf{R}^2)$

The set $f^{-1}(\mathbf{y})$ can be parameterized as $\mathbf{x}_1(\mu; y_1, y_2) = (\mu, \mu - y_2, y_1 + y_2 - 2\mu)^T$ with $\mu \in (-\infty, \infty)$.

$\sqrt{|D_\mu \mathbf{x}_1^T(\mu; \mathbf{y})\ D_\mu \mathbf{x}_1(\mu; \mathbf{y})|} = \|(1, 1, -2)\| = \sqrt{6}$

And, $\sqrt{|Df(\mathbf{x}_1(\mu; \mathbf{y}))\ Df^T(\mathbf{x}_1(\mu; \mathbf{y}))|} = \sqrt{\left| \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 0 \end{pmatrix} \right|} = \sqrt{6}$

Therefore, Theorem 1 gives:

$$P_{Y_1Y_2}(y_1, y_2) = \int_{-\infty}^{\infty} P_{X_1X_2X_3}(\mu, \mu - y_2, y_1 + y_2 - 2\mu)\, d\mu$$

Example 5a: $y_1 = x_1 + x_2 + x_3$; $y_2 = x_1 - x_2 + x_4$ $\quad (f : \mathbf{R}^4 \to \mathbf{R}^2)$

The set $f^{-1}(\mathbf{y})$ can be parameterized as $\mathbf{x}_1(\mu_1, \mu_2; y_1, y_2) = (\mu_1, \mu_2, y_1 - \mu_1 - \mu_2, y_2 - \mu_1 + \mu_2)^T$ with $\mu_1, \mu_2 \in (-\infty, \infty)$.

$$\sqrt{|D_\mu \mathbf{x}_1^T(\mu; \mathbf{y})\ D_\mu \mathbf{x}_1(\mu; \mathbf{y})|} = \sqrt{\left| \begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \\ -1 & 1 \end{pmatrix} \right|} = 3$$

And, $\sqrt{|Df(\mathbf{x}_1(\mu; \mathbf{y}))\ Df^T(\mathbf{x}_1(\mu; \mathbf{y}))|} = \sqrt{\left| \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \right|} = 3$

Therefore, Theorem 1 gives:

$$P_{Y_1Y_2}(y_1, y_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} P_{X_1X_2X_3X_4}(\mu_1, \mu_2, y_1 - \mu_1 - \mu_2, y_2 - \mu_1 + \mu_2)\, d\mu_1\, d\mu_2$$

Example 5b: $y_1 = x_1 + x_2 + x_3$; $y_2 = x_1 - x_2 + x_4$ $\quad (f : \mathbf{R}^4 \to \mathbf{R}^2)$

This is the same function as in Example 5a. The difference in this example is how we parameterize the set $f^{-1}(\mathbf{y})$. In this example we parameterize this set in the following way: $\mathbf{x}_1(\mu_1, \mu_2; y_1, y_2) = (2\mu_1, \mu_2, y_1 - 2\mu_1 - \mu_2, y_2 - 2\mu_1 + \mu_2)^T$ with $\mu_1, \mu_2 \in (-\infty, \infty)$.

Continuing as in 5a we have: $\sqrt{|D_\mu \mathbf{x}_1^T(\mu;\mathbf{y})\; D_\mu \mathbf{x}_1(\mu;\mathbf{y})|} = \sqrt{\left|\begin{pmatrix} 2 & 0 & -2 & -2 \\ 0 & 1 & -1 & 1 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 1 \\ -2 & -1 \\ -2 & 1 \end{pmatrix}\right|} = 6$

And, $\sqrt{|Df(\mathbf{x}_1(\mu;\mathbf{y}))\; Df^T(\mathbf{x}_1(\mu;\mathbf{y}))|} = \sqrt{\left|\begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}\right|} = 3$

Therefore, Theorem 1 gives:

$$P_{Y_1Y_2}(y_1,y_2) = 2\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} P_{X_1X_2X_3X_4}(2\mu_1,\mu_2,y_1-2\mu_1-\mu_2,y_2-2\mu_1+\mu_2)\,d\mu_1\,d\mu_2$$

Substituting $s = 2\mu_1$ shows that this equals the result of Example 5a. The two parameterizations cover the same manifold, and the surface-measure factor $\sqrt{|D_\mu \mathbf{x}^T D_\mu \mathbf{x}|}$ exactly compensates for the change of parameterization: formula (2) is independent of the choice of charts. The answer belongs to the manifold, not to the coordinates we happen to lay on it – a useful self-test the reader can apply to any computation with formula (2).

Example 6: $y = \sum_{i=1}^n x_i^2 \quad (f : \mathbf{R}^n \to \mathbf{R})$

We assume that the x_i are independent normals with mean 0 and variance σ^2. When $\sigma = 1$, y has a Chi-squared distribution of order n. We derive a generalization of this fact using our formula. The set $f^{-1}(y)$ is a sphere in $\mathbf{R}^n$ of radius $\sqrt{y}$. The density function of y is:[4]

$$P_Y(y) = \int_{\partial B(\sqrt{y})} \frac{P_{\mathbf{X}}(\mathbf{x})\,dS}{\sqrt{|Df(\mathbf{x})\;Df^T(\mathbf{x})|}}$$

Here, $Df(\mathbf{x}) = 2\,\mathbf{x}^T$. Consequently, on the sphere of radius $\sqrt{y}$ we have $\sqrt{|Df(\mathbf{x})Df^T(\mathbf{x})|} = 2\sqrt{y}$. Therefore, we may write the above as

$$P_Y(y) = \int_{\partial B(\sqrt{y})} \frac{P_{\mathbf{X}}(\mathbf{x})\,dS}{2\sqrt{y}}$$

This is

$$P_Y(y) = \int_{\partial B(\sqrt{y})} \frac{e^{-\sum_{i=1}^n x_i^2/(2\sigma^2)}\,dS}{(2\pi)^{n/2}\sigma^n 2\sqrt{y}}$$

Which is

$$P_Y(y) = \int_{\partial B(\sqrt{y})} \frac{e^{-y/(2\sigma^2)}\,dS}{(2\pi)^{n/2}\sigma^n 2\sqrt{y}}$$

Or,

$$P_Y(y) = \frac{e^{-y/(2\sigma^2)}}{(2\pi)^{n/2}\sigma^n 2\sqrt{y}} \int_{\partial B(\sqrt{y})} dS$$

[4] We use the surface-measure form of the transformation formula, equation (1), with $f^{-1}(y)$ the sphere of radius $\sqrt{y}$; no global parameterization of the sphere is needed.

The surface area of a sphere of radius $\sqrt{y}$ in n dimensions is $2y^{(n-1)/2}\frac{\pi^{n/2}}{\Gamma(n/2)}$.[5] Therefore, $P_Y(y)$, may be written:

$$P_Y(y) = \frac{y^{n/2-1}e^{-y/(2\sigma^2)}}{2^{n/2}\sigma^n\Gamma(n/2)} \quad y > 0,$$

with $P_Y(y) = 0$ for $y \leq 0$. When $\sigma = 1$ this is the Chi-squared density with n degrees of freedom, as claimed.

Application to the Distribution of the Ratio of Normals

We use Theorem 1 to compute the distribution of the ratio of two correlated normals.

If x_1, x_2 are Gaussian random variables with joint density function, $P_{X_1X_2}$, means: μ_1, μ_2, and positive definite covariance matrix:[6] $\Sigma = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, then the density function of the ratio $y = x_1/x_2$ is given by the formula in Example 2 (this density was studied classically by Geary [3] and Hinkley [4]):

$$P_Y(y) = \int_{-\infty}^{\infty} |s| P_{X_1X_2}(sy, s)\, ds = \frac{1}{2\pi\sqrt{ac-b^2}} \int_{-\infty}^{\infty} |s| e^{-\left([sy-\mu_1, s-\mu_2]\Sigma^{-1}[sy-\mu_1, s-\mu_2]^T\right)/2}\, ds$$

Writing the exponent as $-(\alpha s^2 - 2\beta s + 2\gamma)/2$, completing the square in s, and using the identity[7]

$$\int_{-\infty}^{\infty} |s|\, e^{-\alpha s^2/2+\beta s}\, ds = \frac{2}{\alpha}\left[1 + \sqrt{\pi}\,\tau\, e^{\tau^2}\operatorname{erf}(\tau)\right], \qquad \tau = \frac{|\beta|}{\sqrt{2\alpha}},$$

this may be reduced to:

$$P_Y(y) = \frac{\sqrt{ac-b^2}e^{-\gamma}}{\pi(cy^2-2by+a)}\left[1 + \sqrt{\pi}\tau(y)e^{\tau(y)^2}\operatorname{erf}\left(\tau(y)\right)\right]$$

where

$$\gamma = \frac{c\mu_1^2 - 2b\mu_1\mu_2 + a\mu_2^2}{2(ac-b^2)}$$
$$\tau(y) = \frac{|y(c\mu_1 - b\mu_2) + a\mu_2 - b\mu_1|}{\sqrt{cy^2-2by+a}\sqrt{2(ac-b^2)}}$$

The function $\tau(y)$ is bounded on the interval $(-\infty, \infty)$: by the positive definiteness of Σ the denominator is bounded away from 0, and as $|y| \to \infty$, $\tau(y) \to |c\mu_1 - b\mu_2|/\sqrt{2c(ac-b^2)}$. A continuous function with finite limits at $\pm\infty$ is bounded.

Note: As a consequence, this distribution has no finite moments of order ≥ 1. This follows since $P_Y(y)$ is $O(1/y^2)$ for large $|y|$, so $\int |y|^k P_Y(y)\, dy$ diverges for every $k \geq 1$. *Therefore, one cannot talk about the mean or standard deviation of this distribution.*

This distribution is also not unimodal in general; Marsaglia [5] gives explicit bimodal examples.

[5] This Γ-function formula for sphere volumes and surface areas is derived in the companion paper "Defining Fractal Dimension".

[6] If Σ is positive definite we have: $a > 0,\ c > 0,\ ac - b^2 > 0$.

[7] The identity is not deep: split the integral at $s = 0$ and fold the negative half onto the positive, giving $2\int_0^\infty s\, e^{-\alpha s^2/2}\cosh(\beta s)\, ds$; then complete the square in each of $e^{\pm\beta s}$ and substitute – the surviving Gaussian integrals produce the erf term.

Special Cases

The density function simplifies under the condition: $\mu_1 = \mu_2 = 0$. $P_Y(y)$ is given by:

$$P_Y(y) = \frac{\sqrt{ac-b^2}}{\pi(cy^2-2by+a)} = \frac{\sqrt{ac-b^2}}{\pi\left((\sqrt{c}y-\frac{b}{\sqrt{c}})^2+(a-\frac{b^2}{c})\right)}$$

If, in addition, x_1 and x_2 are uncorrelated, then $P_Y(y)$ becomes:

$$P_Y(y) = \frac{\sqrt{ac}}{\pi(cy^2+a)}$$

Finally, if we further set, $a = c = 1, b = 0$; that is, x_1 and x_2 are independent Gaussians each with mean 0 and variance 1, then $P_Y(y)$ becomes:

$$P_Y(y) = \frac{1}{\pi(y^2+1)}$$

References

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[5] G. Marsaglia, Ratios of normal variables, *Journal of Statistical Software* **16**(4) (2006), 1–10.